



School of Technology, Design and Computer Application
Silver Oak College of Computer Application
Bachelor of Science Computer Science & Information Technology

Semester:	5th	Academic Year:	2025-26
Course Name:	Introduction to Internet of Things	Course Code:	3040243336

Question bank

Sr. No.	Question Text	Marks	CO
Unit No-1 : Internet of Things Overview			
1	Describe in brief the evolution of IoT technology.	3M	CO1
2	Define the following terms: i) Internet of Things (IoT) ii) Gateway iii) Cloud iv) Edge Computing v) Smart Device	5M	CO1
3	Enlist & describe in brief the characteristics of IoT	4M	CO1
4	Explain in brief key components of IoT.	4M	CO1
5	Illustrate the functioning of an IoT system with any one real world example.	4M	CO1
6	Explain in detail M2M architecture with examples.	5M	CO1
7	Explain in detail End-to-End architecture with examples.	5M	CO1
8	Compare and contrast Machine-to-Machine (M2M) architecture with End-to-End IoT architecture.	5M	CO1
9	Enlist advantages and disadvantages of IoT technology.	4M	CO1
10	Describe the interdependence between IoT and cloud computing	5M	CO1
11	List out the names and role of the programming languages essential for IoT application development at different layers (device, gateway, cloud, and application).	4M	CO1
Unit No-2 : IoT Devices, Embedded Systems and Cloud Integration			
12	Define the following terms: i) Sensors ii) Actuators iii) Transducers iv) Microcontroller v) Microprocessor	3M	CO2
13	Write a brief note on sensors in IoT and explain any two sensors in detail.	4M	CO2

14	Write a brief note on actuators in IoT and explain any two actuators in detail.	4M	CO2
15	Classify the sensors based on output and datatype in brief.	4M	CO2
16	Enlist different types of sensors	3M	CO2
17	Enlist different types of actuators	3M	CO2
18	Describe the following: i) IR Sensor ii) Humidity DHT-11 Sensor iii) Ultrasonic Sensor iv) Moisture Sensor	4M	CO2
19	Describe the following: i) Pneumatic Actuator ii) Electric Actuator	4M	CO2
20	Write the difference between Arduino Uno & ESP32.	4M	CO2
21	Explain the working of ESP32 in brief.	4M	CO2
22	Write a brief note on Raspberry Pi.	4M	CO2
23	Explain in detail Embedded IoT system with neat and clean diagram	5M	CO2
24	Explain how device control through the cloud using mobile & web applications is possible.	5M	CO2
25	List out the names of a few IoT platforms and explain the role in IoT ecosystems.	4M	CO2
26	Evaluate the performance of different cloud storage providers (e.g., AWS, Google Cloud, Azure) for IoT data storage and retrieval.	5M	CO2
Programs:			
27	Write a program in Arduino IDE to blink an LED using ESP32.	3M	CO2
28	Write a program in Arduino IDE to control two LEDs using two push buttons.	4M	CO2
29	Write a program in Arduino IDE to read IR sensor data and display it on the Serial Monitor.	4M	CO2
30	Write a program in Arduino IDE to read temperature and humidity from the DHT11 sensor and display data on the Serial Monitor.	5M	CO2
31	Write a program in Arduino IDE to control an LED using Bluetooth with ESP32.	4M	CO2
32	Write a program in Arduino IDE to interface DHT11 sensor and two LEDs with ESP32 and display/control them using the Blynk App.	5M	CO2
33	Write a Python program to blink an LED using Raspberry Pi.	3M	CO2
34	Write a Python program to simulate a traffic light control using three LEDs using Raspberry Pi.	5M	CO2

Unit No-3 : IoT Protocols, Networks and Security Challenges			
35	Define IoT communication protocols. And list-out the different layers of the IoT communication protocol stack.	4M	CO3
36	Explain few Link layer/Physical Layer protocols in brief and also enlist its real-world examples.	4M	CO3
37	Explain few Network layer protocols in brief and also enlist its real-world examples.	4M	CO3
38	Explain few Transport layer protocols in brief and also enlist its real-world examples.	4M	CO3
39	Explain few Application layer protocols in brief and also enlist its real-world examples.	4M	CO3
40	Explain how the MQTT protocol facilitates communication in IoT.	4M	CO3
41	Write the difference between MQTT and HTTP.	4M	CO3
42	Explain in detail following protocols: i) MQTT ii) CoAP iii) XMPP	3M	CO3
43	Classify different types of sensor network topologies and describe each with a neat and clean diagram.	5M	CO3
44	Describe briefly the security components for IoT systems.	4M	CO3
45	Discuss in detail about the challenges in IoT security.	4M	CO3
Unit No-4 : IoT Applications and Entrepreneurship			
46	Draw and explain common architecture for IoT applications.	5M	CO4
47	Illustrate any two IoT applications in detail for home automation system.	4M	CO4
48	How would you integrate IoT into an agriculture system to monitor soil moisture levels and automate irrigation? Describe the hardware and software components for the same.	5M	CO4
49	Discuss IoT application as health care monitoring. (Or) Explain how IoT enhances healthcare systems. Use examples to illustrate the advantages.	5M	CO4
50	Illustrate in detail any two IoT applications that can be implemented in Smart Cities	4M	CO4
51	Explain the role of Artificial Intelligence in enhancing IoT applications. Give suitable examples.	5M	CO4
52	List and briefly describe the key challenges faced by IoT-based startups.	4M	CO4



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53	Design a step-by-step process to transform an IoT idea into a functional prototype.	5M	CO4
54	Enlist different business models for IoT ventures.	3M	CO4
55	Explain any two business models commonly adopted by IoT ventures.	4M	CO4

Course Coordinator

Head of Department